Configuring Link Aggregation to Improve Bandwidth

Difficulty (out of 3 stars): ★★★ Recommended study time: 1 hour

Contents

1 About This Lab

1.1 Task Description

As networks grow in scale, users require backbone links to provide higher bandwidth and reliability. In conventional technologies, a common practice to increase the bandwidth is to use high-speed LPUs or devices that support high-speed LPUs, which is costly and inflexible.

In comparison, link aggregation bundles multiple physical interfaces into a logical interface to increase the link bandwidth without upgrading hardware. In addition, link aggregation provides link backup mechanisms, greatly improving link reliability. It has the following advantages:

- Increased bandwidth: The maximum bandwidth of an Eth-Trunk interface is the sum of bandwidth of its member interfaces.
- Higher reliability: When an active link fails, traffic on this link is switched to another active link, enhancing reliability.
- Load balancing: Traffic is load balanced among all available active links.

After completing this task, you will be able to configure Huawei switches, understand link aggregation, and know how to configure link aggregation on switches.

1.2 Objectives

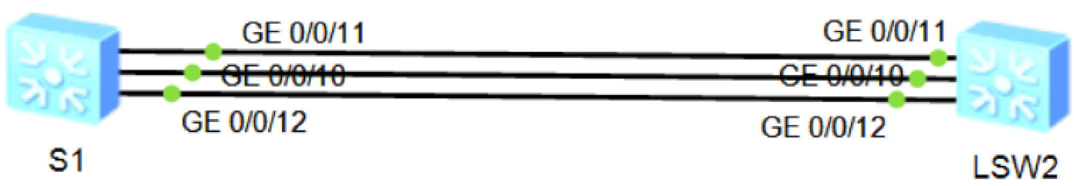
On completing this task, you will be able to:

1. Learn how to manually configure link aggregation.
2. Learn how to configure link aggregation in static LACP mode.
3. Learn how to determine active links in static LACP mode.
4. Learn how to configure some static LACP features.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology



In the spanning tree lab, the two links between S1 and S2 cannot be both in the data forwarding state at the same time. To make full use of the bandwidth of the two links, configure Ethernet link aggregation between S1 and S2.

2.2 Initial Configuration

- Initial configuration of S1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S1
[S1]
```

- Initial configuration of S2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S2
[S2]
```

3 Task Implementation

3.1 Checking Basic Configurations

Omitted

3.2 Configuring Link Aggregation in Manual Mode

1. Create an Eth-Trunk interface.

```
[S1]interface Eth-Trunk 1
```

The **interface eth-trunk** command displays the view of an existing Eth-Trunk or creates an Eth-Trunk and displays its view. The number **1** in this example indicates an interface ID. The interface ID range varies according to devices.

```
[S2]interface Eth-Trunk 1
```

Configure the link aggregation mode of the Eth-Trunk interface.

```
[S1-Eth-Trunk1]mode manual load-balance
```

The **mode** command configures the working mode of the Eth-Trunk interface, which can be LACP or manual load balancing. By default, the manual load balancing mode is used. Therefore, the preceding operation is unnecessary and is provided for demonstration purpose only.

2. Add member interfaces to an Eth-Trunk interface.

```
[S1]interface GigabitEthernet 0/0/10
[S1-GigabitEthernet0/0/10]eth-trunk 1
Info: This operation may take a few seconds. Please wait for a moment...done.
[S1-GigabitEthernet0/0/10]quit
[S1]interface GigabitEthernet 0/0/11
[S1-GigabitEthernet0/0/11]eth-trunk 1
Info: This operation may take a few seconds. Please wait for a moment...done.
[S1-GigabitEthernet0/0/11]quit
[S1]interface GigabitEthernet 0/0/12
[S1-GigabitEthernet0/0/12]eth-trunk 1
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S1-GigabitEthernet0/0/12]quit
```

You can enter the interface view of an individual interface and add it to an Eth-Trunk interface. You can also run the **trunkport** command in the Eth-Trunk interface view to add multiple interfaces to the Eth-Trunk interface at a batch.

```
[S2]interface Eth-Trunk 1
[S2-Eth-Trunk1]trunkport GigabitEthernet 0/0/10 to 0/0/12
Info: This operation may take a few seconds. Please wait for a moment...done.
```

Note the following points when adding interfaces to an Eth-Trunk interface:

- An Eth-Trunk interface supports up to 8 member interfaces.
- An Eth-Trunk interface cannot be added to another Eth-Trunk interface as a member interface.
- An Ethernet interface can be added to only one Eth-Trunk interface. To add the Ethernet interface to another Eth-Trunk interface, delete it from the current Eth-Trunk interface first.
- If an interface of the local device is added to an Eth-Trunk interface, an interface of the remote device directly connected to the interface of the local device must also be added to an Eth-Trunk interface so that the two ends can communicate.
- Both ends of an Eth-Trunk must use the same number of physical interfaces, rate, and duplex mode.

3. Check the status of the Eth-Trunk interface.

```
[S1]display eth-trunk 1
Eth-Trunk1's state information is:
WorkingMode: NORMAL              Hash arithmetic: According to SIP-XOR-DIP
Least Active-linknumber: 1       Max Bandwidth-affected-linknumber: 32
Operate status: up               Number Of Up Port In Trunk: 3
--------------------------------------------------------------------------------
PortName                        Status         Weight
GigabitEthernet0/0/10           Up             1
GigabitEthernet0/0/11           Up             1
GigabitEthernet0/0/12           Up             1
```

3.3 Configuring Link Aggregation in LACP Mode

1. Delete member interfaces from an Eth-Trunk interface.

```
[S1]interface Eth-Trunk 1
[S1-Eth-Trunk1]undo trunkport GigabitEthernet 0/0/10 to 0/0/12
Info: This operation may take a few seconds. Please wait for a moment...done.

[S2]interface Eth-Trunk 1
[S2-Eth-Trunk1]undo trunkport GigabitEthernet 0/0/10 to 0/0/12
Info: This operation may take a few seconds. Please wait for a moment...done.
```

Before changing the working mode of an Eth-Trunk interface, ensure that the Eth-Trunk interface has no member interfaces.

2. Change the working mode of the Eth-Trunk interface.

```
[S1]interface Eth-Trunk 1
[S1-Eth-Trunk1]mode lacp
```

The **mode lacp** command sets the working mode of an Eth-Trunk interface to LACP. Note: The command is **mode lacp-static** in some versions.

```
[S2]interface Eth-Trunk 1
[S2-Eth-Trunk1]mode lacp
```

3. Add member interfaces to an Eth-Trunk interface.

```
[S1]interface Eth-Trunk 1
[S1-Eth-Trunk1]trunkport GigabitEthernet 0/0/10 to 0/0/12
Info: This operation may take a few seconds. Please wait for a moment...done.

[S2]interface Eth-Trunk 1
[S2-Eth-Trunk1]trunkport GigabitEthernet 0/0/10 to 0/0/12
Info: This operation may take a few seconds. Please wait for a moment...done.
```

4. Check the status of the Eth-Trunk interface.

```
[S1]display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                          WorkingMode: STATIC
Preempt Delay: Disabled            Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768             System ID: 4c1f-cc33-7359
Least Active-linknumber: 1         Max Active-linknumber: 8
Operate status: up                 Number Of Up Port In Trunk: 3
--------------------------------------------------------------------------------
ActorPortName          Status    PortType  PortPri PortNo PortKey PortState Weight
GigabitEthernet0/0/10  Selected  1GE       32768   11     305     10111100  1
GigabitEthernet0/0/11  Selected  1GE       32768   12     305     10111100  1
GigabitEthernet0/0/12  Selected  1GE       32768   13     305     10111100  1

Partner:
--------------------------------------------------------------------------------
ActorPortName          SysPri    SystemID        PortPri  PortNo  PortKey  PortState
GigabitEthernet0/0/10  32768     4c1f-ccc1-4a02  32768    11      305      10111100
GigabitEthernet0/0/11  32768     4c1f-ccc1-4a02  32768    12      305      10111100
GigabitEthernet0/0/12  32768     4c1f-ccc1-4a02  32768    13      305      10111100
```

3.4 Considering Network Traffic

In normal cases, only GigabitEthernet0/0/11 and GigabitEthernet0/0/12 are in the forwarding state, and GigabitEthernet0/0/10 serves as a backup. When the number of active interfaces falls below 2, the Eth-Trunk interface is shut down.

1. Set the LACP priority of S1 to have S1 become the Actor.

```
[S1]lacp priority 100
```

2. Configure interface priorities so that GigabitEthernet0/0/11 and GigabitEthernet0/0/12 can have a higher priority.

```
[S1]interface GigabitEthernet 0/0/10
[S1-GigabitEthernet0/0/10]lacp priority 40000
```

LACPDUs are sent and received by devices at both ends of a LAG in LACP mode.

First, the Actor is elected.

- The system priority field is compared. The default priority value is 32768, and a smaller value indicates a higher priority. The device with a higher priority is elected as the LACP Actor.

- In case of the same system priority, the device with a smaller MAC address becomes the Actor.
- After the Actor is elected, the devices at both ends select active interfaces according to the interface priority settings on the Actor. A smaller interface priority value indicates a higher priority. The default priority value is 32768.

3. Set the upper and lower thresholds for the number of active interfaces in an Eth-Trunk interface.

```
[S1]interface Eth-Trunk 1
[S1-Eth-Trunk1]max active-linknumber 2
[S1-Eth-Trunk1]least active-linknumber 2
```

The bandwidth and status of an Eth-Trunk interface depend on the number of active interfaces in the Eth-Trunk interface. The bandwidth of an Eth-Trunk is the total bandwidth of all member interfaces in the Up state. You can set the following thresholds to stabilize an Eth-Trunk's status and bandwidth as well as reduce the impact brought by frequent changes of member link status.

- Lower threshold: When the number of active interfaces falls below this threshold, the Eth-Trunk interface goes Down. This threshold determines the minimum bandwidth of an Eth-Trunk interface. The **least active-linknumber** command sets the lower threshold for the number of active interfaces in an Eth-Trunk interface.
- Upper threshold: When the number of active interfaces reaches this threshold, the bandwidth of the Eth-Trunk interface will not increase even if more member links go Up. The upper threshold improves network reliability while assuring bandwidth. The **max active-linknumber** command sets the upper threshold for the number of active interfaces in an Eth-Trunk interface.

4. Enable the preemption function.

```
[S1]interface Eth-Trunk 1
[S1-Eth-Trunk1]lacp preempt enable
```

In LACP mode, when an active link fails, the system selects the standby link with the highest priority to replace the faulty link. If the faulty link recovers and has a higher priority than the standby link and the LACP priority preemption function is enabled, the link with a higher priority replaces the link with a lower priority as an active link. The **lacp preempt enable** command enables LACP preemption. By default, this function is disabled.

5. Check the status of the Eth-Trunk interface.

```
[S1]display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                          WorkingMode: STATIC
Preempt Delay Time: 30             Hash arithmetic: According to SIP-XOR-DIP
System Priority: 100               System ID: 4c1f-cc33-7359
Least Active-linknumber: 2         Max Active-linknumber: 2
Operate status: up                 Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
ActorPortName          Status     PortType  PortPri  PortNo  PortKey  PortState  Weight
GigabitEthernet0/0/10  Unselect   1GE       40000    11      305      10100000   1
GigabitEthernet0/0/11  Selected   1GE       32768    12      305      10111100   1
GigabitEthernet0/0/12  Selected   1GE       32768    13      305      10111100   1

Partner:
--------------------------------------------------------------------------------
ActorPortName          SysPri     SystemID       PortPri  PortNo  PortKey  PortState
```

```
GigabitEthernet0/0/10    32768       4c1f-ccc1-4a02  32768      11         305          10110000
GigabitEthernet0/0/11    32768       4c1f-ccc1-4a02  32768      12         305          10111100
GigabitEthernet0/0/12    32768       4c1f-ccc1-4a02  32768      13         305          10111100
```

GigabitEthernet0/0/11 and GigabitEthernet0/0/12 are in the active state.

6. Shut down GigabitEthernet0/0/12 to simulate a link fault.

```
[S1]interface GigabitEthernet 0/0/12
[S1-GigabitEthernet0/0/12]shutdown
```

```
[S1]display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                          WorkingMode: STATIC
Preempt Delay Time: 30             Hash arithmetic: According to SIP-XOR-DIP
System Priority: 100              System ID: 4c1f-cc33-7359
Least Active-linknumber: 2       Max Active-linknumber: 2
Operate status: up                Number Of Up Port In Trunk: 2
--------------------------------------------------------------------------------
ActorPortName            Status     PortType    PortPri PortNo    PortKey PortState Weight
GigabitEthernet0/0/10     Selected    1GE        40000      11        305          10111100    1
GigabitEthernet0/0/11     Selected    1GE        32768      12        305          10111100    1
GigabitEthernet0/0/12     Unselect    1GE         32768      13        305          10100010    1

Partner:
--------------------------------------------------------------------------------
ActorPortName             SysPri      SystemID         PortPri    PortNo    PortKey PortState
GigabitEthernet0/0/10     32768       4c1f-ccc1-4a02   32768       11         305          10111100
GigabitEthernet0/0/11     32768       4c1f-ccc1-4a02   32768       12         305          10111100
GigabitEthernet0/0/12     0           0000-0000-0000 0           0          0            10100011
```

GigabitEthernet 0/0/10 has become active.

7. Shut down GigabitEthernet 0/0/11 to simulate a link fault.

```
[S1]interface GigabitEthernet 0/0/11
[S1-GigabitEthernet0/0/11]shutdown
```

```
[S1]display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                          WorkingMode: STATIC
Preempt Delay Time: 30             Hash arithmetic: According to SIP-XOR-DIP
System Priority: 100              System ID: 4c1f-cc33-7359
Least Active-linknumber: 2       Max Active-linknumber: 2
Operate status: down               Number Of Up Port In Trunk: 0
--------------------------------------------------------------------------------
ActorPortName            Status     PortType   PortPri   PortNo    PortKey    PortState Weight
GigabitEthernet0/0/10     Unselect 1GE         40000       11         305         10100000    1
GigabitEthernet0/0/11     Unselect 1GE         32768       12         305         10100010    1
GigabitEthernet0/0/12     Unselect 1GE         32768       13         305         10100010    1

Partner:
--------------------------------------------------------------------------------
```

```
ActorPortName          SysPri    SystemID        PortPri  PortNo  PortKey  PortState
GigabitEthernet0/0/10  32768     4c1f-ccc1-4a02  32768    11      305      10110000
GigabitEthernet0/0/11  0         0000-0000-0000  0        0       0        10100011
GigabitEthernet0/0/12  0         0000-0000-0000  0        0       0        10100011
```

The lower threshold for the number of active interfaces in the Eth-Trunk interface is set to 2. Therefore, when the number of active interfaces in the Eth-Trunk interface is less than 2, all the member interfaces are shut down. Although GigabitEthernet0/0/10 is Up, it is still in the Unselect state.

3.5 Changing the Load Balancing Mode

1. Enable the interfaces that are shut down in the previous step.

```
[S1]inter GigabitEthernet 0/0/11
[S1-GigabitEthernet0/0/11]undo shutdown
[S1-GigabitEthernet0/0/11]quit
[S1]inter GigabitEthernet 0/0/12
[S1-GigabitEthernet0/0/12]undo shutdown
```

2. Wait about 30 seconds and check the status of Eth-Trunk 1.

```
[S1]display eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                     WorkingMode: STATIC
Preempt Delay Time: 30        Hash arithmetic: According to SIP-XOR-DIP
System Priority: 100          System ID: 4c1f-cc33-7359
Least Active-linknumber: 2    Max Active-linknumber: 2
Operate status: down          Number Of Up Port In Trunk: 0
--------------------------------------------------------------------------------
ActorPortName          Status    PortType  PortPri PortNo  PortKey  PortState  Weight
GigabitEthernet0/0/10  Unselect  1GE       40000   11      305      10100000   1
GigabitEthernet0/0/11  Selected  1GE       32768   12      305      10100010   1
GigabitEthernet0/0/12  Selected  1GE       32768   13      305      10100010   1

Partner:
--------------------------------------------------------------------------------
ActorPortName          SysPri    SystemID        PortPri  PortNo  PortKey  PortState
GigabitEthernet0/0/10  32768     4c1f-ccc1-4a02  32768    11      305      10110000
GigabitEthernet0/0/11  0         0000-0000-0000  0        0       0        10100011
GigabitEthernet0/0/12  0         0000-0000-0000  0        0       0        10100011
```

The preemption function is enabled on the Eth-Trunk interface. Therefore, when GigabitEthernet0/0/11 and GigabitEthernet0/0/12 enter the Up state, the two interfaces have a higher priority than GigabitEthernet0/0/10. As a result, GigabitEthernet0/0/10 enters the Unselect state. The system waits 30 seconds by default before allowing preemption to maintain link stability. Preemption starts only after this delay when the interfaces become active.

3. Change the load balancing mode of the Eth-Trunk interface to destination IP address-based load balancing.

```
[S1]interface Eth-Trunk 1
[S1-Eth-Trunk1]load-balance dst-ip
```

To ensure proper load balancing between physical links of an Eth-Trunk interface and avoid link congestion, run the **load-balance** command to set the load balancing mode of the Eth-

Trunk interface. Load balancing is valid only for outgoing traffic; therefore, the load balancing modes on the interfaces at both ends of an Eth-Trunk interface can be different and do not affect each other.

4 Verification

Omitted.

5 Quiz

1. What are the requirements for the values of **least active-linknumber** and **max active-linknumber?**
2. What are the differences between packet-based load balancing and flow-based load balancing?

   Packet-based load balancing distributes packets to different links, which may cause packet disorder. Flow-based load balancing distributes packets of the same flow to the same link, which prevents packet disorder. However, a single flow cannot use the logical bandwidth of the entire aggregation interface.
3. How is the Actor elected in LACP mode?

   The Actor is elected by comparing system priorities. A smaller value indicates a higher priority. If the system priorities are the same, the bridge MAC addresses are compared. A smaller value indicates a higher priority. The device with the highest priority becomes the Actor.